Filtering joins

JOINING DATA WITH PANDAS

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Mutating versus filtering joins

Mutating joins:

- Combines data from two tables based on matching observations in both tables

Filtering joins:

- Filter observations from table based on whether or not they match an observation in another table

What is a semi join?

Left Table

A	B	C
A2	B2	C2
A3	B3	C3
A4	B4	C4

Right Table

C	D
C1	D1
C2	D2
C4	D4
C5	D5

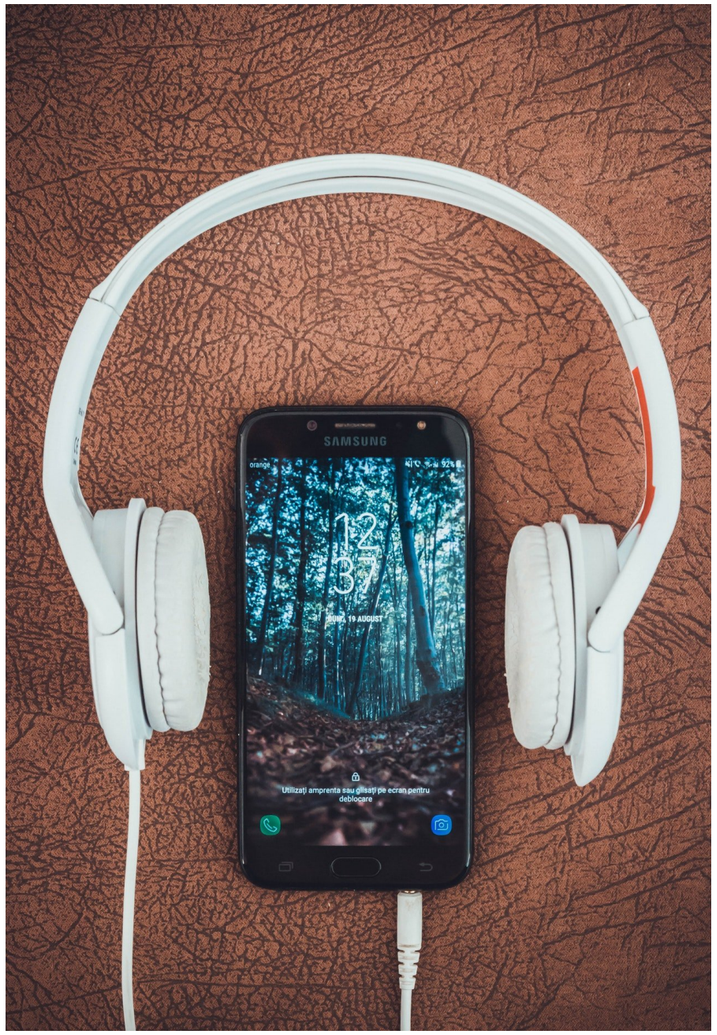
Result Table

A	B	C
A2	B2	C2
A4	B4	C4

Semi joins

- Returns the intersection, similar to an inner join
- Returns only columns from the left table and ***not*** the right
- No duplicates

Musical dataset



SQLITE TUTORIAL

[1] Photo by Vlad Bagacian from Pexels

Example datasets

```
  gid  name
0 1    Rock
1 2    Jazz
2 3    Metal
3 4    Alternative ...
4 5    Rock And Roll
```

```
  tid  name             aid  mtid  gid  composer         u_price
0 1    For Those Ab...  1    1     1    Angus Young,...  0.99
1 2    Balls to the...  2    2     1    nan              0.99
2 3    Fast As a Shark  3    2     1    F. Baltes, S...  0.99
3 4    Restless and...  3    2     1    F. Baltes, R...  0.99
4 5    Princess of ...  3    2     1    Deaffy & R.A...  0.99
```

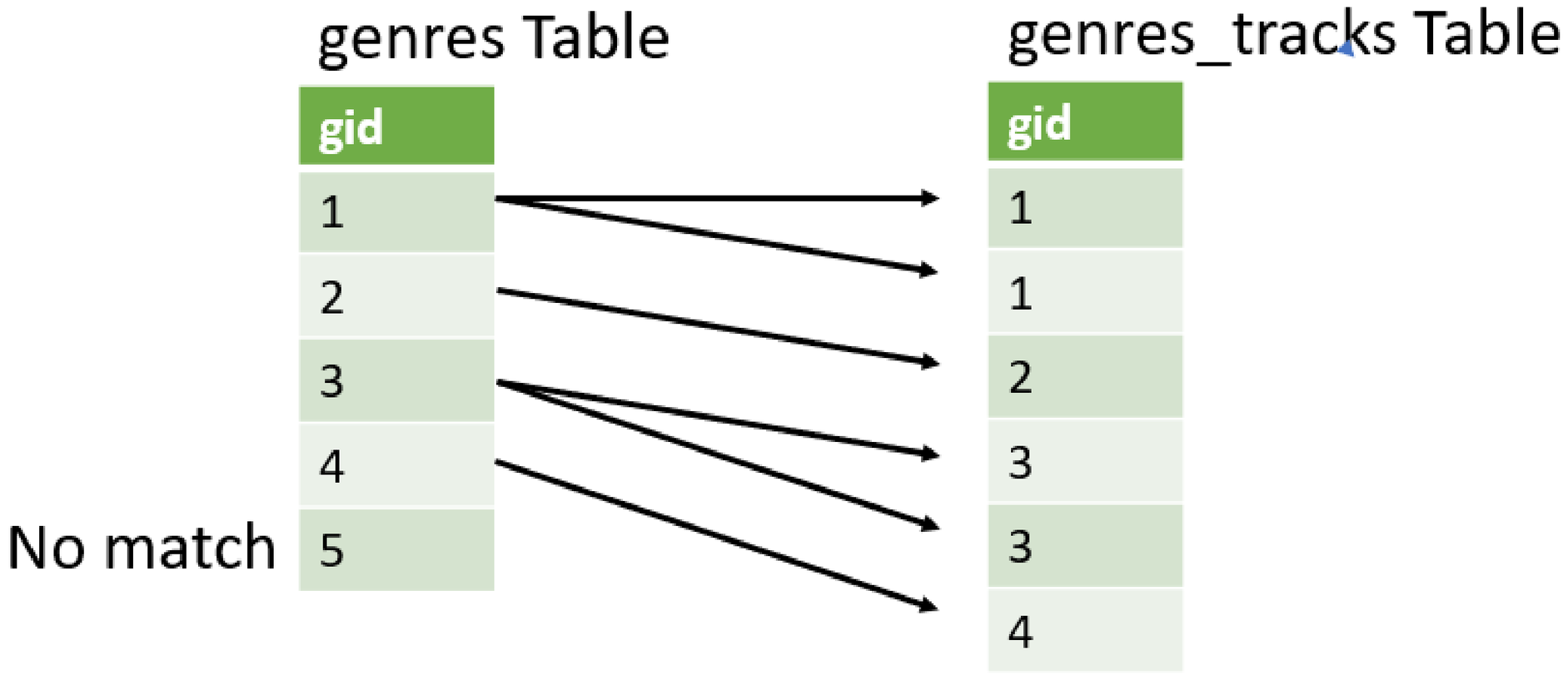
Step 1 - semi join

```
genres_tracks = genres.merge(top_tracks, on='gid')
print(genres_tracks.head())
```

```
  gid  name_x  tid   name_y           aid  mtid  composer         u_price
0 1    Rock    2260  Don't Stop M...  185  1     Mercury, Fre...  0.99
1 1    Rock    2933  Mysterious Ways  232  1     U2               0.99
2 1    Rock    2618  Speed Of Light   212  1     Billy Duffy/...  0.99
3 1    Rock    2998  When Love Co...  237  1     Bono/Clayton...  0.99
4 1    Rock    685   Who'll Stop ...  54   1     J. C. Fogerty    0.99
```

Step 2 - semi join

```
genres['gid'].isin(genres_tracks['gid'])
```



Step 2 - semi join

```
genres['gid'].isin(genres_tracks['gid'])
```

```
0      True
1      True
2      True
3      True
4     False
Name: gid, dtype: bool
```

Step 3 - semi join

```
genres_tracks = genres.merge(top_tracks, on='gid')
top_genres = genres[genres['gid'].isin(genres_tracks['gid'])]
print(top_genres.head())
```

```
  gid  name
0 1    Rock
1 2    Jazz
2 3    Metal
3 4    Alternative & Punk
4 6    Blues
```

What is an anti join?

Left Table

A	B	C
A2	B2	C2
A3	B3	C3
A4	B4	C4

Right Table

C	D
C1	D1
C2	D2
C4	D4
C5	D5

Result Table

A	B	C
A3	B3	C3

Anti join:

- Returns the left table, excluding the intersection
- Returns only columns from the left table and ***not*** the right

Step 1 - anti join

```
genres_tracks = genres.merge(top_tracks, on='gid', how='left', indicator=True)
print(genres_tracks.head())
```

```
  gid  name_x          tid     name_y           aid    mtid  composer         u_price  _merge
0 1    Rock            2260.0  Don't Stop M...  185.0  1.0   Mercury, Fre...  0.99     both
1 1    Rock            2933.0  Mysterious Ways  232.0  1.0   U2               0.99     both
2 1    Rock            2618.0  Speed Of Light   212.0  1.0   Billy Duffy/...  0.99     both
3 1    Rock            2998.0  When Love Co...  237.0  1.0   Bono/Clayton...  0.99     both
4 5    Rock And Roll   NaN     NaN              NaN    NaN   NaN              NaN      left_only
```

Step 2 - anti join

```
gid_list = genres_tracks.loc[genres_tracks['_merge'] == 'left_only', 'gid']
print(gid_list.head())
```

```
23     5
34     9
36    11
37    12
38    13
Name: gid, dtype: int64
```

Step 3 - anti join

```
genres_tracks = genres.merge(top_tracks, on='gid', how='left', indicator=True)
gid_list = genres_tracks.loc[genres_tracks['_merge'] == 'left_only','gid']
non_top_genres = genres[genres['gid'].isin(gid_list)]
print(non_top_genres.head())
```

```
  gid  name
0 5    Rock And Roll
1 9    Pop
2 11   Bossa Nova
3 12   Easy Listening
4 13   Heavy Metal
```

Let's practice!

JOINING DATA WITH PANDAS

Concatenate DataFrames together vertically

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Concatenate two tables vertically

A	B	C
A1	B1	C1
A2	B2	C2
A3	B3	C3

A	B	C
A4	B4	C4
A5	B5	C5
A6	B6	C6

- pandas `.concat()` method can concatenate both vertical and horizontal.
  - `axis=0`, vertical

Basic concatenation

- 3 different tables
- Same column names
- Table variable names:
  - `inv_jan` *(top)*
  - `inv_feb` *(middle)*
  - `inv_mar` *(bottom)*

```
  iid  cid  invoice_date  total
0 1    2    2009-01-01    1.98
1 2    4    2009-01-02    3.96
2 3    8    2009-01-03    5.94
```

```
  iid  cid  invoice_date  total
0 7    38   2009-02-01    1.98
1 8    40   2009-02-01    1.98
2 9    42   2009-02-02    3.96
```

```
  iid  cid  invoice_date  total
0 14   17   2009-03-04    1.98
1 15   19   2009-03-04    1.98
2 16   21   2009-03-05    3.96
```

Basic concatenation

```
pd.concat([inv_jan, inv_feb, inv_mar])
```

```
   iid  cid  invoice_date  total
0  1    2    2009-01-01    1.98
1  2    4    2009-01-02    3.96
2  3    8    2009-01-03    5.94
0  7    38   2009-02-01    1.98
1  8    40   2009-02-01    1.98
2  9    42   2009-02-02    3.96
0  14   17   2009-03-04    1.98
1  15   19   2009-03-04    1.98
2  16   21   2009-03-05    3.96
```

Ignoring the index

```
pd.concat([inv_jan, inv_feb, inv_mar],
          ignore_index=True)
```

	iid	cid	invoice_date	total
0	1	2	2009-01-01	1.98
1	2	4	2009-01-02	3.96
2	3	8	2009-01-03	5.94
3	7	38	2009-02-01	1.98
4	8	40	2009-02-01	1.98
5	9	42	2009-02-02	3.96
6	14	17	2009-03-04	1.98
7	15	19	2009-03-04	1.98
8	16	21	2009-03-05	3.96

Setting labels to original tables

```
pd.concat([inv_jan, inv_feb, inv_mar],
          ignore_index=False,
          keys=['jan','feb','mar'])
```

```
        iid  cid  invoice_date  total
jan 0   1    2    2009-01-01    1.98
    1   2    4    2009-01-02    3.96
    2   3    8    2009-01-03    5.94
feb 0   7    38   2009-02-01    1.98
    1   8    40   2009-02-01    1.98
    2   9    42   2009-02-02    3.96
mar 0   14   17   2009-03-04    1.98
    1   15   19   2009-03-04    1.98
    2   16   21   2009-03-05    3.96
```

Concatenate tables with different column names

Table: `inv_jan`

```
  iid  cid  invoice_date  total
0 1    2    2009-01-01    1.98
1 2    4    2009-01-02    3.96
2 3    8    2009-01-03    5.94
```

Table: `inv_feb`

```
  iid  cid  invoice_date  total  bill_ctry
0 7    38   2009-02-01    1.98   Germany
1 8    40   2009-02-01    1.98   France
2 9    42   2009-02-02    3.96   France
```

Concatenate tables with different column names

```
pd.concat([inv_jan, inv_feb],
          sort=True)
```

```
  bill_ctry  cid  iid  invoice_date  total
0 NaN        2    1    2009-01-01    1.98
1 NaN        4    2    2009-01-02    3.96
2 NaN        8    3    2009-01-03    5.94
0 Germany    38   7    2009-02-01    1.98
1 France     40   8    2009-02-01    1.98
2 France     42   9    2009-02-02    3.96
```

Concatenate tables with different column names

```
pd.concat([inv_jan, inv_feb],
          join='inner')
```

iid	cid	invoice_date	total
1	2	2009-01-01	1.98
2	4	2009-01-02	3.96
3	8	2009-01-03	5.94
7	38	2009-02-01	1.98
8	40	2009-02-01	1.98
9	42	2009-02-02	3.96

Let's practice!

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Verifying integrity

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Let's check our data

Possible merging issue:

A	B	C
A1	B1	C1
A2	B2	C2
A3	B3	C3

C	D
C1	D1
C1	D2
C1	D3
C2	D4

- Unintentional one-to-many relationship
- Unintentional many-to-many relationship

Possible concatenating issue:

A	B	C
A1	B1	C1
A2	B2	C2
A3	B3	C3

A	B	C
A3 (duplicate)	B3 (duplicate)	C3 (duplicate)
A4	B4	C4
A5	B5	C5

- Duplicate records possibly unintentionally introduced

Validating merges

`.merge(validate=None)`:

- Checks if merge is of specified type
- `'one_to_one'`
- `'one_to_many'`
- `'many_to_one'`
- `'many_to_many'`

Merge dataset for example

Table Name: `tracks`

```
  tid  name              aid  mtid  gid  u_price
0 2    Balls to the...   2    2     1    0.99
1 3    Fast As a Shark   3    2     1    0.99
2 4    Restless and...   3    2     1    0.99
```

Table Name: `specs`

```
  tid  milliseconds  bytes
0 2    342562        5510424
1 3    230619        3990994
2 2    252051        4331779
```

Merge validate: one_to_one

```
tracks.merge(specs, on='tid',
             validate='one_to_one')
```

```
Traceback (most recent call last):
MergeError: Merge keys are not unique in right dataset; not a one-to-one merge
```

Merge validate: one_to_many

```
albums.merge(tracks, on='aid',
             validate='one_to_many')
```

```
  aid  title            artid  tid  name            mtid  gid  u_price
0 2    Balls to the...  2      2    Balls to the...  2     1    0.99
1 3    Restless and...  2      3    Fast As a Shark  2     1    0.99
2 3    Restless and...  2      4    Restless and...  2     1    0.99
```

Verifying concatenations

`.concat(verify_integrity=False)`:

- Check whether the new concatenated index contains duplicates
- Default value is `False`

Dataset for .concat() example

Table Name: `inv_feb`

```
     cid  invoice_date  total
iid
7    38   2009-02-01    1.98
8    40   2009-02-01    1.98
9    42   2009-02-02    3.96
```

Table Name: `inv_mar`

```
     cid  invoice_date  total
iid
9    17   2009-03-04    1.98
15   19   2009-03-04    1.98
16   21   2009-03-05    3.96
```

Verifying concatenation: example

```
pd.concat([inv_feb, inv_mar],
          verify_integrity=True)
```

```
Traceback (most recent call last):
ValueError: Indexes have overlapping
values: Int64Index([9], dtype='int64',
name='iid')
```

```
pd.concat([inv_feb, inv_mar],
          verify_integrity=False)
```

```
     cid  invoice_date  total
iid
7    38   2009-02-01    1.98
8    40   2009-02-01    1.98
9    42   2009-02-02    3.96
9    17   2009-03-04    1.98
15   19   2009-03-04    1.98
16   21   2009-03-05    3.96
```

Why verify integrity and what to do

Why:

- Real world data is often ***NOT*** clean

What to do:

- Fix incorrect data
- Drop duplicate rows

Let's practice!

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